1A

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:126:45: warning: '.tmp' directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncati
on=]
  126 |     snprintf(tmp_path, sizeof(tmp_path), "%s.tmp", path);
      |                                             ^~~~
object.c:126:5: note: 'snprintf' output between 5 and 516 bytes into a destination of size 512
  126 |     snprintf(tmp_path, sizeof(tmp_path), "%s.tmp", path);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$
```

1B

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$
```

2A

```
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ xxd .pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442 | head -20
00000000: 7472 6565 2034 3900 3130 3036 3434 2068  tree 49.100644 h
00000010: 656c 6c6f 2e74 7874 0066 2246 63d2 3e6f  ello.txt.f"Fc.>o
00000020: 4d9d e9e2 c7e6 d876 4305 a92a 3830 a1a5  M......vC..*80..
00000030: 2d3d 5f4a a800 7b5c 39                   -=_J..{\9
```

3A

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ nano pes.c
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ make pes
rm -rf .pes
./pes init
echo "hello world" > file1.txt
echo "goodbye world" > file2.txt
./pes add file1.txt file2.txt
./pes status
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
    staged: file1.txt
    staged: file2.txt

Unstaged changes:
    (nothing to show)

Untracked files:
    untracked: test_objects.c
    untracked: pes.c
    untracked: .gitignore
    untracked: commit.h
    untracked: index.h
    untracked: Makefile
    untracked: test_objects
    untracked: tree.h
    untracked: tree.c
    untracked: pes.h
    untracked: test_sequence.sh
    untracked: index.c
    untracked: object.c
    untracked: commit.c
    untracked: test_tree.c
    untracked: README.md
    untracked: test_tree
```

3B

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 17766575
26 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 17766575
26 6 file2.txt
```

4A

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ rm -rf .pes
./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
Initialized empty PES repository in .pes/
Committed: fa613ed26c19... Initial commit
Committed: 8e35895174b3... Add world
Committed: 410ec4125d09... Add farewell
commit 410ec4125d0930d125ee2af3766700d955a138e570a0b6e4e8e80725fb2b9431
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658467

    Add farewell

commit 8e35895174b3b4eeddb33a3e3a15d92b2e9fe6bcb6679dacd8181904137e2b6d
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658467

    Add world

commit fa613ed26c196a40b341c5c5743f664c717712805365d9b2addb30895a1a028f
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658467

    Initial commit
```

4B

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/41/0ec4125d0930d125ee2af3766700d955a138e570a0b6e4e8e80725fb2b9431
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/8e/35895174b3b4eeddb33a3e3a15d92b2e9fe6bcb6679dacd8181904137e2b6d
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/fa/613ed26c196a40b341c5c5743f664c717712805365d9b2addb30895a1a028f
.pes/refs/heads/main
```

4C

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
410ec4125d0930d125ee2af3766700d955a138e570a0b6e4e8e80725fb2b9431
ref: refs/heads/main
```

make test-integration

```
bhavyaa@debian:~/PES1UG24CS116-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
    staged: file.txt
    staged: hello.txt

Unstaged changes:
    (nothing to show)

Untracked files:
    (nothing to show)


--- First Commit ---
Committed: d1791a275437... Initial commit

Log after first commit:
commit d1791a2754377dc450041246a5a7f73acbd2a67409d545d9a09146128411b8d6
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658869

    Initial commit
```

```
--- Second Commit ---
Committed: 5da627a64613... Update file.txt

--- Third Commit ---
Committed: f81fe1e17ff3... Add farewell

--- Full History ---
commit f81fe1e17ff30991d111a9797315e843bf46c7d51d762766e1af6a30c6d7be17
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658869

    Add farewell

commit 5da627a64613a3d6c6f44c2c7dfb7e52d9d82fd54f2b774b06958f1ad24ef54e
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658869

    Update file.txt

commit d1791a2754377dc450041246a5a7f73acbd2a67409d545d9a09146128411b8d6
Author: Bhavyaa Garg <PES1UG24CS116>
Date:   1776658869

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
f81fe1e17ff30991d111a9797315e843bf46c7d51d762766e1af6a30c6d7be17
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/5d/a627a64613a3d6c6f44c2c7dfb7e52d9d82fd54f2b774b06958f1ad24ef54e
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/d1/791a2754377dc450041246a5a7f73acbd2a67409d545d9a09146128411b8d6
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f8/1fe1e17ff30991d111a9797315e843bf46c7d51d762766e1af6a30c6d7be17

=== All integration tests completed ===
```